MATTEO PAOLINI

Mechanical Design, Mechatronics & Robotics

Brussels, Belgium | EU citizen | Open to relocation
Available from February 2027
paolini134@gmail.com | Portfolio | linkedin.com/in/matpaolini

EDUCATION

M.Sc. Mechanical Engineering — Mechatronics 2024–Feb 2027 (expected)
Università Politecnica delle Marche
Erasmus coursework at Université libre de Bruxelles and Vrije Universiteit Brussel, 2025–2026.

B.Sc. Mechanical Engineering 2021–2024
Università Politecnica delle Marche
Thesis: collision-free motion planning for the 7-DOF TIAGo arm using MATLAB, including an RRT parameter study and path shortcutting.

ENGINEERING EXPERIENCE & PROJECTS

Royal Military Academy — Robot Learning for Humanitarian Demining Jul 2026–Present
Research Internship | Brussels, Belgium

- Developing a hierarchical policy that looks at the scene and decides *what* to remove and *how*, to safely expose a landmine: a PPO-trained decider, a skill manager that selects skills or skill combinations, and a push skill, trained with teacher–student distillation.
- Wrote the robot software for a UR5e arm with Robotiq 2F-140 gripper as independent modules (scene, simulation, perception, decision, execution, task) with typed interfaces; only the simulation module touches MuJoCo, so the real robot can replace it.
- Pipeline: training in mjlab (MuJoCo), validation with ROS in Gazebo, then deployment on the real robot.

Apple Watch Case — Reverse Engineering, Design & Production Study 2025
Academic Project | UNIVPM | Team of two

- Reverse-engineered the case in Siemens NX from public drawings and teardown images; selected Al 7005-T6 and assigned functional GD&T, including ±0.01 mm on precision fits and Ø0.05 mm concentricity on the Ø3 H7 crown bore.
- Defined the manufacturing route: laser blanking (608 blanks per panel), CNC cavity and contour milling with a custom vise, and vibratory finishing.
- Sized a theoretical production line for 1,000,000 parts per year from material-removal rates and cutting power (1 laser cutter, 89 CNC centers, 1 tumbler), resolved the modeled milling bottleneck by scheduling the fleet over four periods, and built tool-life (Taylor) and cost-per-part models.

Pneumatic Soft Gripper Sep 2025–Jan 2026
Academic project | VUB/ULB | Team of five

- Designed the tentacle, holder and casting mold with a removable core in Siemens NX; 3D-printed the components and casting tooling. Traced inconsistent bending to an off-center mold core that varied wall thickness along the part.
- Compared internal chamber geometries using FEA in ANSYS Workbench to guide the tentacle's bending response, using a second-order Ogden hyperelastic model.
- Designed tentacles for quick replacement and easy mounting on a holder accommodating different tentacle types and sizes; integrated the gripper with the team on a working 2-DOF robot.

Data-Driven Fault Diagnostics — Motor Bearings & Multiphase Plant 2026
Academic projects | UNIVPM

- Re-tested a published bearing-fault classifier on motor-current data (99.6% reported): my reconstruction reached 46.9%. Traced the gap to the output layer's range (minimum −1.76 vs. currents near −3 A), so the residual measured clipping; removing it raised AUC from 0.60 to 0.77.
- Multiphase plant: found all fault recordings came from one operating point; designed leakage-free temporal splits; a regime-conditioned Random Forest detected 502–515 of 523 faults with zero false alarms (global model: 49–58).

Also: dual-arm handover simulation (2 Franka Panda arms, Drake). **Details and code:** matpaol.github.io

ADDITIONAL INDUSTRIAL EXPERIENCE

Machinery Maintenance Assistant — Seasonal | PM Macchine Summers 2016–2020
Corropoli, Italy | Occasional support ongoing

- Maintained industrial leather-goods machinery alongside experienced technicians and independently: sewing-machine hook timing, pneumatic troubleshooting, and replacement of hydraulic seals, sensors and electronic control boards.

Assembly & Machine Calibration Intern | Co.Me.Ta. Srl 2019
Montegranaro, Italy

- Assembled frames, integrated sensors, motors and electromechanical components, and performed final functional calibration.

Assembly & Wiring Intern | Galli S.p.A. 2018
Vigevano, Italy

- Assembled and wired industrial machinery for leather-goods production.

TECHNICAL SKILLS

Mechanical design: Siemens NX | SolidWorks | Engineering drawings | GD&T | DFM | Process planning
Simulation & analysis: Numerical simulation | Rigid-body dynamics | ANSYS Workbench | Hyperelastic FEA | MATLAB/Simulink | MuJoCo | mjlab | ROS/Gazebo
Prototyping & machinery: FDM 3D printing | Mold design | Silicone casting | Assembly & wiring | Pneumatic troubleshooting | Hydraulic maintenance
Robotics & programming: Python | PyTorch | PPO | Teacher–student RL | Stable-Baselines3 | Drake | Arduino | Git
Languages: Italian — native | English — C1